# Title:
# Healthcare Data Analysis: Diabetes Risk Assessment

**Prepared by:**

**Md Zulkar Nain**

**Role:**

**Aspiring Data Analyst**

**Tools Used:**

Python, Pandas, NumPy, Matplotlib, Seaborn

Dataset Source:

Kaggle

**Date:**

Feb 2026

## Executive Summary

This project analyzes a healthcare dataset to identify key factors associated with diabetes risk. The analysis focuses on data cleaning, exploratory data analysis, and statistical comparison between diabetic and non-diabetic patients. Results indicate that glucose levels, BMI, and age are significant indicators of diabetes risk. These insights can help healthcare professionals prioritize early screening and preventive care.

# Problem Statement

Diabetes is one of the most common chronic diseases worldwide. Early identification of risk factors can help in prevention and early treatment.

The goal of this project is to analyze patient health metrics to identify key factors that are associated with diabetes risk and extract meaningful insights from the dataset.

## Dataset Description

- Dataset contains **768 patient records**
- **8 medical attributes + 1 target variable**
- Target variable: **Outcome**
    - 0 = Non-diabetic
    - 1 = Diabetic

# Dataset Features

| Feature | Description |
|---|---|
| Pregnancies | Number of pregnancies |
| Glucose | Blood glucose concentration |
| BloodPressure | Blood pressure measurement |
| SkinThickness | Skin fold thickness |
| Insulin | Serum insulin level |
| BMI | Body Mass Index |
| DiabetesPedigreeFunction | Genetic influence factor |
| Age | Age of patient |
| Outcome | Diabetes status |

## Tools and Technologies

Python

Pandas

NumPy

Matplotlib

Seaborn

Jupyter Notebook

## Data Cleaning

The dataset contained invalid zero values in several medical attributes where zero is not medically possible.

The following preprocessing steps were performed:

- Replaced zero values in **Glucose, BMI, BloodPressure, Insulin, and SkinThickness** using median imputation.
- Checked for duplicate records and removed them.
- Verified data consistency using summary statistics.

Why median?

Median was chosen because medical datasets often contain skewed distributions.

## Exploratory Data Analysis (EDA)

Include 3–4 visualizations such as:

- Age distribution
- BMI distribution
- Glucose comparison by outcome
- Correlation heatmap

The glucose distribution indicates significantly higher values among diabetic patients compared to non-diabetic patients.

## Key Findings

- The dataset shows moderate class imbalance (approx. 65% non-diabetic vs 35% diabetic). This should be considered during model development to avoid biased predictions The probability of diabetes increases

- Although diabetic patients show significantly higher median glucose levels, the distributions overlap, indicating that glucose alone is insufficient for definitive classification and should be evaluated alongside other health indicators.
- Diabetic patients exhibit approximately 28–29% higher average glucose levels compared to non-diabetic patients, indicating glucose as a strong distinguishing health indicator.
- Approximately 35% of patients in the dataset are diabetic, indicating a significant minority class that should be carefully analyzed for risk factor identification.

## Conclusion:

The analysis indicates that glucose level is the strongest indicator of diabetes risk, followed by BMI and age. These findings highlight the importance of regular glucose monitoring and lifestyle management to reduce diabetes risk.